

Which Shadows the Substrate Casts: a Coherent/Disorder Decomposition of Event Density's Continuum Limits

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Series: Event Density (ED) Generative Papers — substrate-evaluation arc (the continuum-limit question, decomposition method) **Status:** Publication draft. Empirical substrate-evaluation with the certified Σ -rule simulator (**Bits/**). Standalone; cold-reader accessible. Companion to *The Continuum Limit of the Certified ED Substrate* (kinetic-lattice-gas) and *From Worldlines to the Canonical PDE* (channel ledger). **Repository target:** physics-papers/substrate-evaluation/ (ED-Generative)

Preamble: What This Paper Does NOT Claim

1. **It does not prove Event Density.** The result is consilience evidence — one discrete substrate reproducing, or failing to reproduce, several known continuum laws under coarse-graining. Consilience accumulates; it is not a single decisive experiment, and the paper is explicit about what is recovered, what is recovered as a *different* object, and what is not recovered at all.
 2. **The Maxwell recovery is partly expected by construction.** Once the substrate is identified as a lattice gauge theory (Paper_MS-I), a Maxwell continuum limit is what such theories generically produce; the result confirms that the substrate has the right structure, not that Maxwell arises from nothing.
 3. **The 13 substrate primitives are not derived** (position paper, Paper_087). These are results about the *certified Σ -substrate*, not a derivation of ED.
 4. **Single substrate instantiation (the Σ -rule grid), modest sizes, single-configuration sweeps.** Absolute fit values are not the result; the *contrasts* are (a coherent field that tracks Coulomb vs a total that does not; transport beating diffusion; a phase-randomized surrogate that is Gaussian while the field is not).
 5. **“Continuum law” means the coarse-grained / projected description, not a claim about fundamental law.** The substrate is the fundamental object here; the continuum equations are its shadows, in the sense made precise below.
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Abstract

Coarse-graining a discrete substrate raises a question sharper than “does it give the continuum?”: *which* continuum law does it cast, and is that law the textbook object one expected? We coarse-grain the certified Event-Density substrate and read off the answer with a single decomposition — separating the **coherent** part of a field (its ensemble-consistent signal) from the **incoherent** part (the realization-specific disorder) — applied to three continuum questions. The three answers are genuinely different. (i) **Charge $\rightarrow$ Maxwell, a window:** the coherent part of the coarse gauge field tracks the Coulomb (action-minimizing) field, with the incoherence localized at the source as entropy; the unseparated total does not, because it sums field and entropy. (ii) **Diffusion $\rightarrow$ transport, a different object:** the ensemble-mean density obeys a

clean continuum law, but the eikonal/transport equation (regression R^2 0.60), not diffusion (R^2 0.26) — Event Density coarse-grains to the *hyperbolic* pole, not the diffusive one. (iii) **Gaussianity** → **a wall**: a phase-randomized surrogate carrying the substrate’s exact power spectrum is essentially perfectly Gaussian (skewness 0.003), while the substrate field is strongly non-Gaussian (skewness 0.96) — so 100% of the non-Gaussianity lives in the Fourier *phases*, which is to say in the substrate’s coherent structure; there is no Gaussian signal to recover. The discriminator across the three is one question — *is there a coherent signal under the disorder, and is it the object asked for?* — and it characterizes the substrate in one line: **Event Density is a structure-making, hyperbolic-pole, phase-correlating substrate**. That phrase predicts which continuum laws of the PDE Atlas it casts by building (the hyperbolic and structure-making ones) and which it does not reach by building (the diffusive and structureless ones), giving a falsifiable map rather than a blanket claim.

1. Introduction

A substrate ontology earns the word “shadow” for the continuum only if it can show, concretely, which continuum laws its own dynamics cast and which they do not. The companion kinetic-lattice-gas paper showed the certified Event-Density substrate’s coarse density field is a kinetic (transport) equation, not a diffusion PDE. This paper generalizes the question from one law to many, and sharpens it: rather than asking “does the substrate give the continuum,” we ask of each candidate law,

which coarse-graining floor casts this shadow, and does the substrate’s averaging climb high enough to cast it?

We answer it with a single tool — a **coherent/incoherent decomposition** — applied to three continuum questions (Maxwell, diffusion, Gaussianity). The three come back different: a window (the textbook object is recovered), a different object (a clean law, but not the one expected), and a wall (no coherent object to recover). The pattern across the three is itself the result: a one-line characterization of the substrate that predicts, for the broader catalogue of continuum equations, which it casts and which live a coarse-graining floor up.

2. Background

The substrate. The certified Σ -rule (Bits/) evolves an active front on a participation graph: it scores admissible neighbors by Σ = coherence – strain – gradient, commits at the winner (the arrow), deposits density, and advances. It is local, generative, and committal.

The decomposition. A field measured from the substrate is a sum of an **ensemble-coherent** part — the structure that is consistent across realizations or commit orders — and an **incoherent** part — the realization-specific disorder. Most “does it give the continuum” tests measure a *quadratic* quantity (an energy, a variance, a cumulant) that mixes the two; the decomposition separates them, so that the coherent signal can be compared to the candidate continuum law directly and the disorder can be identified for what it is.

The pole taxonomy. The continuum laws fall into structural classes (the PDE Atlas): hyperbolic (transport, Hamilton–Jacobi, Burgers), diffusive (porous-medium, Fokker–Planck, reaction–diffusion), dispersive, geometric, and so on. A law’s pole is a property of the *shadow*; the question here is which poles the substrate casts.

3. Method

For each continuum question we coarse-grain the substrate, separate the coherent signal from the incoherent disorder, and apply one discriminator:

- **is there a coherent signal under the disorder, and is it the object asked for?**

Three outcomes are possible: (a) the coherent part *is* the textbook object — a **window**; (b) the coherent part is a clean continuum law but a *different* object — a **different object**; (c) there is no coherent object — a **wall**. The three tests:

- **Charge $\rightarrow$ Maxwell:** ensemble-average the gauge field as the coherent quantity $\hat{e}^{\{i\}}$ (not the energy), and compare the coherent-field deficit to the action-minimizing (Coulomb) reference; identify the incoherence radially.
- **Diffusion:** ensemble-average the density to extract the coherent (mean) field, and regress its time-derivative on a candidate library (diffusion ², eikonal/transport $| \cdot |$).
- **Gaussianity:** build the phase-randomized surrogate — the substrate’s *exact* power spectrum with randomized Fourier phases — and compare its real-space cumulants to the field’s; this isolates whether the non-Gaussianity lives in the phases.

4. Results

4.1 Charge $\rightarrow$ Maxwell: a window

Charge in this substrate is an integer winding of the participation phase (Paper_ChargeAsTopology_B4); the Maxwell action is latent in the coherence term, which penalizes phase gradients as $\frac{1}{4}(\cdot)^2$. Measuring the *total* coarse deficit (the energy) yields a profile that grows with radius and does not match Coulomb — because the energy sums the coherent field and the incoherent disorder. Separating them changes the picture. The **coherent-field** deficit $\cdot r^2$ (ensemble of 384, $L = 121$) is

r	4	8	16	32	48
coherent field	0.285	0.196	0.169	0.164	0.153
Maxwell (action- minimizing) reference	0.301	0.240	0.245	0.190	0.151

The coherent field tracks the Maxwell reference, matching it at the far edge (0.153 vs 0.151), and it cleans up monotonically as the ensemble grows. The **incoherence** $1 - |\hat{e}^{\{i\}}|$ is sharply localized at the source (0.92 at $r = 4$) and falls toward zero far away (0.14 at $r = 48$), and the coherence magnitude plateaus near 0.67 across ensemble sizes — a stable disorder fraction, not a finite-sample effect. The coarse-grained substrate casts **Maxwell in the coherent part, with the entropy localized at the charge**. (Note item 2: this confirms the lattice-gauge structure produces its expected continuum limit.)

4.2 Diffusion $\rightarrow$ transport: a different object

Ensemble-averaging the density extracts a smooth coherent field whose evolution is well-described by a single continuum law — but not diffusion. For a step initial condition, regressing the mean field’s time-derivative gives

model	single realization R^2	ensemble-mean R^2
diffusion (²)	0.001	0.258
eikonal / transport ($\cdot$)	0.094	0.597

Ensemble-averaging cleans the dynamics dramatically (R^2 0.10 $\rightarrow$ 0.60), confirming a clean continuum law emerges; that law is **eikonal/transport**, which beats diffusion decisively. The mean density spreads ballistically ($\propto t$), not diffusively ($\propto \sqrt{t}$), because the substrate’s worldlines are straight rather than random-walk. The coarse-grained substrate casts the **hyperbolic-pole** shadow; the diffusive pole lives a coarse-graining floor up, where the mixing the substrate does not perform would be manufactured.

4.3 Gaussianity $\rightarrow$ a wall

A Gaussian random field is defined by independent, random Fourier phases; amplitude-Gaussianity is uninformative, since each Fourier mode is a sum over all of space and is therefore near-Gaussian for almost any field. The discriminating test is the phase-randomized surrogate — the field’s exact power spectrum with phases randomized:

	skewness	excess kurtosis
substrate field	0.958	−0.325
phase-randomized surrogate	0.003	−0.001

The surrogate — same amplitudes, randomized phases — is essentially perfectly Gaussian, so the power spectrum carries none of the non-Gaussianity. The substrate field, with its actual phases, is strongly non-Gaussian. **100% of the non-Gaussianity lives in the phases** (97% at a coarser scale). Correlated phases are precisely what structure is — a filament is a phase-aligned object — so the substrate is non-Gaussian for the same reason it is committal: it builds coherent structure rather than mixing it away. There is no Gaussian signal to recover; this is a wall, not a masked window.

5. Synthesis

5.1 The discriminator and the three outcomes

question	coherent part	outcome
Charge $\rightarrow$ Maxwell	= the textbook object (Coulomb field)	window
Diffusion	a clean continuum law, but a <i>different</i> object (transport/eikonal)	different object
Gaussianity	<i>no</i> coherent object (the field is intrinsically non-Gaussian)	wall

The decomposition reveals *what continuum law the substrate actually casts*, and it is not always the textbook object one asked for, and sometimes there is no clean object at all. The discriminator — *is there a coherent signal under the disorder, and is it the object asked for?* — is the content, and it is sharper than a blanket “does it give the continuum.”

5.2 The characterization, and the map it predicts

The three outcomes have one source. The substrate **makes structure** (its non-Gaussianity is entirely phase correlation), it is **hyperbolic** (its coarse density transports rather than diffuses), and it is **phase-correlating** (the coherent part is where its content lives). In one line:

Event Density is a structure-making, hyperbolic-pole, phase-correlating substrate.

That phrase predicts the Atlas. The substrate casts, *by building*, the structure-making and hyperbolic shadows — transport, Hamilton–Jacobi, Burgers, and (through its lattice-gauge structure) Maxwell. It does not reach, *by building*, the structureless and mixing shadows — the diffusive pole and Gaussianity — because building coherent structure is the opposite of the mixing those require; they live a coarse-graining floor up, reached only when a mixing or dissipative ingredient is added (the companion channel-ledger paper identifies those ingredients as real but switched-off ED primitives). The prediction is falsifiable: a structure-making substrate that cast a clean Gaussian field, or whose coarse density diffused on its own, would refute the characterization.

6. Discussion

The form of the evidence. A claim about a hidden substrate is confirmed the way atomism was — by consilience, one substrate accounting for many independent known laws without retuning between them. The value of the present result is not any single recovery but the *pattern* and its *discriminator*: a substrate that casts the hyperbolic and structure-making shadows, casts no Gaussian one, and tells you in advance which is which. The Maxwell recovery is partly expected once the lattice-gauge structure is granted (item 2); the transport result is the substrate confirming its own ballistic character; the Gaussianity wall is a genuine negative. A negative, located precisely (the non-Gaussianity is the phase structure), is as informative as a positive.

Relation to the companion papers. The kinetic-lattice-gas paper established the transport (not diffusion) result for the coarse *field*; this paper recovers it as one of three outcomes and adds the Maxwell window and the Gaussianity wall. The channel-ledger paper explains *why* the diffusive and dissipative shadows are not cast by the bare rule — they require ingredients (mixing, capacity, global feedback) the minimal substrate lacks — which is the mechanism behind this paper’s “floor up.”

Limitations. Single substrate instantiation, modest sizes, single-configuration sweeps; the contrasts, not the absolute fit values, carry the conclusions. The Maxwell coherent-field tracks the action-minimizing reference closely but, on a finite box with a finite source, neither reaches the asymptotic Coulomb constant; the claimed equivalence is coherent-field = action-minimizer, which holds. The transport regression is cleanest for a sharp-front initial condition. The Gaussianity result is decisive but concerns the participation field of this substrate, not a cosmological observable.

7. Conclusion

Coarse-graining the Event-Density substrate and separating coherent signal from incoherent disorder gives, across three continuum questions, three different answers: Maxwell is cast (a window), diffusion is cast as transport (a different object), and Gaussianity is not cast at all (a wall). The discriminator behind the three — *is there a coherent signal under the disorder, and is it the object asked for?* — characterizes the substrate as structure-making, hyperbolic, and phase-correlating, and that characterization predicts which continuum laws of the Atlas it casts and which live a floor up. The picture is a map with its first entries filled in, including one that reads “not here.”